Task 1: List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

Feature	IPv4 Addressing	IPv6 Addressing
Address Size & Space	32-bit address space, allowing approximately 4.3 billion (2^32) unique addresses.	128-bit address space, allowing virtually unlimited (3.4 x 10^38 or 2^128) unique addresses.
Notation Format	Written in dotted-decimal format (e.g., 192.168.10.1) split into four 8-bit octets.	Written in hexadecimal notation separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
Configuration Type	Relies heavily on manual configuration or DHCP (Dynamic Host Configuration Protocol).	Supports stateless auto-configuration (SLAAC) natively, alongside DHCPv6.
Security Protocols	IPSec (IP Security) is optional and must be supported or configured additionally.	IPSec support is built-in and was originally designed to be mandatory for all implementations.

Task 2: Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Scenario: Assigning static IPv4 configurations for 4 personal home devices i would choose class C IP address range eg.- 192.168.10.0/24 network block. Typically, 192.168.10.1 is reserved as the Default Gateway (router).

Device Name	Static IP Address	Subnet Mask	Default Gateway
Personal Laptop	192.168.10.10	255.255.255.0 (/24)	192.168.10.1
Smartphone	192.168.10.11	255.255.255.0 (/24)	192.168.10.1
Smart TV	192.168.10.120	255.255.255.0 (/24)	192.168.10.1
Network Printer	192.168.10.200	255.255.255.0 (/24)	192.168.10.1

Task 3: Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Using a /27 subnet mask on the 192.168.1.0 base network.

• Subnet Mask Decimal: 255.255.255.224
• Number of Subnets: 2^N = 2^3 = 8 Subnets
• Total Hosts per Subnet: = 2^H = 2^5 = 32 Hosts
• Usable Hosts per Subnet: 32 - 2 = 30 Usable Hosts

Subnet ID	Network Address	First Usable IP	Last Usable IP	Broadcast Address
Subnet 1	192.168.1.0	192.168.1.1	192.168.1.30	192.168.1.31
Subnet 2	192.168.1.32	192.168.1.33	192.168.1.62	192.168.1.63
Subnet 3	192.168.1.64	192.168.1.65	192.168.1.94	192.168.1.95

Task 4: You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet. Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

Given IP Address: 10.0.5.17
Subnet Mask: 255.255.255.0

Convert Subnet Mask to CIDR notation:

- 255.255.255.0 = 1111111.1111111.1111111.000000

Network bits= 24

- CIDR notation prefix: /24

Determine Network Boundaries:

- Network Address: 10.0.5.0
- Broadcast Address: 10.0.5.255
- Valid Usable Host Range: 10.0.5.1 to 10.0.5.254

Task 5: Use an online subnet calculator tool (such as https://www.subnet-calculator.com) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.